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A patch that imparts unconditional stability to explicit integrators for Langevin-like equations

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ABSTRACT

This paper extends the results in [8] to stochastic differential equations (SDEs) arising in molecular dynamics. It implements a patch to explicit integrators that consists of a Metropolis–Hastings step. The 'patched integrator' preserves the SDE's equilibrium distribution and is accurate on finite time intervals. As a corollary this paper proves the integrator's accuracy in estimating finite-time dynamics along an infinitely long solution - a first in molecular dynamics. The paper also covers multiple time-steps, holonomic constraints and scalability. Finally, the paper provides numerical tests supporting the theory.

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1. Introduction and motivation

Numerical stability is a necessary condition for convergence of an approximation scheme, and a main theme of this article. This article analyzes the stability of explicit integrators for stochastic differential equations (SDEs) arising in molecular dynamics. The true solution of such SDEs not only exists for all time, but is ergodic with respect to a probability distribution supported at all energies. However, explicit integrators for such systems are stochastically unstable due to nonlinear molecular interactions. This divergence is an established problem for SDEs that possess nonlinear drifts [23,37,47].

This shortcoming of explicit integrators is a consequence of their *conditional stability*. For any time-step there exists an energy above which an explicit integrator is numerically unstable. In this situation stochastic effects necessarily induce instabilities by driving the discretization to energies where the approximation to the drift is repulsive and destabilizing. Since molecular simulations often involve unbounded potential energy (e.g. Lennard–Jones interaction), such energies are attainable.

To be concrete consider the Langevin equations for an 'impact oscillator' [28]. The potential energy function is $U(q) = q^2 + 1/q^2$. It consists of an attractive quadratic part and a repulsive part, and possesses a global minimum at $q = 1$. The governing SDE is given by:

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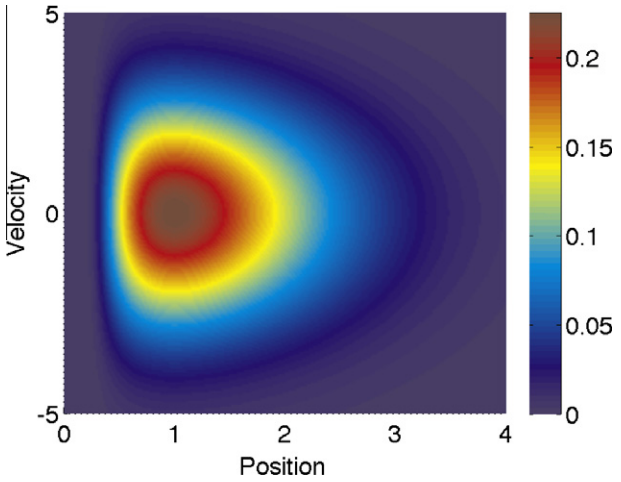


Fig. 1. Contours of SDE's equilibrium density. This figure plots contours of the equilibrium density $\pi(q,v) = Z^{-1}\exp\left(-\beta\left(\frac{v^2}{2}+U(q)\right)\right)$ for the impact oscillator example at inverse temperature $\beta = 0.25$. Observe that its maximum is located at $q = 1$ and $v = 0$.

$$\begin{cases} \frac{dQ}{dt} = V, \\ dV = -\nabla U(Q)\,dt - \gamma V\,dt + \sqrt{2\beta^{-1}\gamma}\,dW, \end{cases} \tag{1}$$

where W is a one-dimensional Brownian motion, γ is a 'thermostat parameter', and β is the 'inverse temperature.' An infinitely long solution of this SDE is distributed according to a probability measure with density π satisfying:

$$\pi(q,v) = C^{-1}\exp\left(-\beta\left(\frac{v^2}{2}+U(q)\right)\right), \quad q>0, \quad v\in\mathbb{R}, \tag{2}$$

where C is a normalization constant. Contours of this equilibrium density are plotted in Fig. 1 for $\beta = 1/4$. Note that π achieves a maximum at $q = 1$ and $v = 0$.

Because of the nonlinear repulsive force, an explicit discretization of (1) is only conditionally stable. For example, consider the following integrator for (1):

$$\begin{cases} V_k^{1/2} = V_k - \frac{h}{2}\nabla U(Q_k), \\ Q_{k+1} = Q_k + hV_k^{1/2}, \\ V_{k+1} = V_k^{1/2} - \frac{h}{2}\nabla U(Q_{k+1}) - \gamma h V_k + \sqrt{2\beta^{-1}\gamma}(W(t_{k+1}) - W(t_k)). \end{cases} \tag{3}$$

This approximation requires roughly the same computational effort as a forward Euler scheme, namely a single potential force evaluation per time-step. It reduces to the classical Verlet algorithm when $\gamma = 0$ [53], and is first-order accurate. In Fig. 2, discrete orbits of the Verlet integrator ($\gamma = 0$) are plotted. Notice that these orbits lose compactness outside a neighborhood of the point $q = 1$ and $v = 0$. When γ is nonzero, the noise can drive the approximation to enter regions where the drift is destabilizing as illustrated by the divergent numerical orbit in Fig. 3. This issue motivates the analysis in [8] and its extension in this paper to SDEs in molecular dynamics.

The issue discussed in this simple example is generic and also arises for SDEs in molecular dynamics which – in addition to nonlinear drifts – are characterized by degenerate noise, high-dimensionality, and multiple time-scales. In this context a main aim of numerical methods is to estimate long-time dynamical properties. This calculation is typically done by launching a single run of an explicit integrator and collecting statistics. However, without the patch described below this approach is prone to failure due to numerical instabilities as illustrated in the preceding impact oscillator example.

In molecular dynamics SDEs arise as descriptions of molecular systems at constant temperature, as we briefly recall here. Consider n molecules with masses m_i for $i = 1,\ldots,n$ evolving in a d-dimensional periodic box (or torus in $\nu = dn$ dimensions: $\mathbb{T}^\nu$). Let $\boldsymbol{M}$ represent the diagonal mass matrix of the molecular system. We assume the particle interaction is derivable from a potential energy function $U:\mathbb{T}^\nu\to\mathbb{R}$. The Hamiltonian $H:\mathbb{T}^\nu\times\mathbb{R}^\nu\to\mathbb{R}$ of this system can be written as:

$$H(\boldsymbol{q},\boldsymbol{p}) = \frac{1}{2}\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} + U(\boldsymbol{q}), \tag{4}$$

where $\boldsymbol{q}\in\mathbb{T}^\nu$ and $\boldsymbol{p}\in\mathbb{R}^\nu$ represent respectively the positions and momenta of the molecules. In terms of this Hamiltonian, define the probability distribution μ:

$$\mu(d\boldsymbol{q},d\boldsymbol{p}) \overset{\text{def}}{=} Z^{-1}e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}d\boldsymbol{q}d\boldsymbol{p}, \quad Z = \int_{\mathbb{T}^\nu\times\mathbb{R}^\nu} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\,d\boldsymbol{q}\,d\boldsymbol{p}. \tag{5}$$

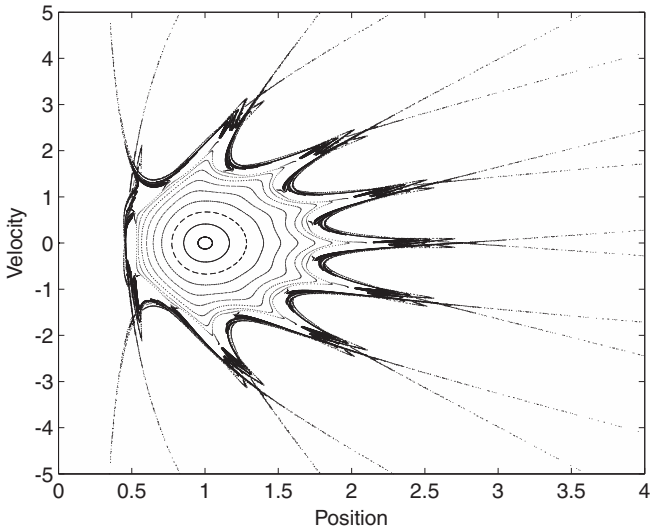


Fig. 2. Conditional stability of Verlet integrator. This figure shows a phase portrait of the Verlet scheme for time-step $h = 0.25$ applied to the impact oscillator example. Notice that orbits in a neighborhood of the fixed point $q = 1$ and $v = 0$ are compact, but compactness is absent in orbits further away. This phase portrait illustrates that the Verlet integrator is only conditionally stable.

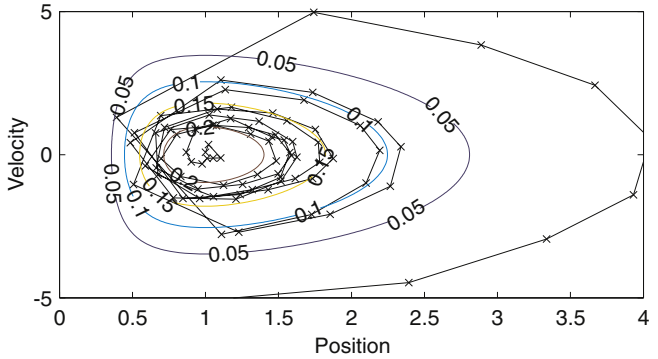


Fig. 3. Exploding orbit of Langevin integrator. This figure illustrates that points along an orbit of the Langevin integrator (3) are not ergodic with respect to the target distribution (2). The orbit is obtained for the impact oscillator example at inverse temperature $\beta = 0.25$, damping parameter $\gamma = 0.0125$ (low damping), time-step $h = 0.25$, and initial condition $q = 1$ and $v = 0$. In the background a few contours of the probability density π are plotted and labeled. The 'x' marks represent points along the orbit. The orbit initially stays nearby its starting point, but escapes the view shown at around $t = 20$.

Here we have introduced the parameter β which is inversely related to the temperature $\mathcal{T}$ and Boltzmann constant k_B via $\beta = 1/(k_B\mathcal{T})$.

A molecular system with Hamiltonian (4) is at constant temperature $\mathcal{T}$ if its trajectories sample from the probability distribution (5). The standard way to guarantee that this is the case is to assume that the molecular system follows a continuous, stochastic dynamics of the form:

$$\begin{cases} \frac{d\boldsymbol{Q}}{dt} = \boldsymbol{M}^{-1}\boldsymbol{P}, \\ d\boldsymbol{P} = -\nabla U(\boldsymbol{Q})\,dt + d\boldsymbol{\eta}(\boldsymbol{Q},\boldsymbol{P}), \end{cases} \tag{6}$$

where $d\boldsymbol{\eta} : \mathbb{T}^\nu \times \mathbb{R}^\nu \to \mathbb{R}^\nu$ represents a thermostat force. Physically one can interpret the thermostat force as modeling interaction between the molecular system and a heat bath. Mathematically the thermostat force ensures that the dynamics of (6) is ergodic with respect to (5).

Several types of thermostat forces are available in the literature [9,10,27,41,43]. They all preserve the probability distribution (5), but which one best captures the dynamics of the molecular system is open for debate. This modeling issue is beyond the scope of the present paper. Here we assume that the thermostat force is given, and we design an integration strategy that does not make strong assumptions on its precise form. In the applications and theory sections of this paper we focus on $d\boldsymbol{\eta}$ given by Langevin dynamics [9,43]. The assumption of continuity excludes, e.g. the Andersen thermostat because it involves discrete collisions at random times that in their wake leave the momentum of the molecular system dis-

continuous [4]. We refer the reader to [14,32] for recent progress quantifying the mixing properties of the Andersen thermostat for molecular systems.

Consider computing the time-correlation in momentum along an infinitely long solution of the SDE. This computation is quite common in molecular dynamics and the reader is referred to [3,18] for expository accounts. Define the continuous equilibrium correlation in momentum as:

$$A(\tau) = \langle \boldsymbol{P}(\tau + t)^T \boldsymbol{P}(t) \rangle \quad (t \geqslant 0) \quad (\tau \geqslant 0), \tag{7}$$

where the angle brackets denote a double average with respect to realizations of (6) and an initial condition distributed according to (5). A commonly used estimate for $A(\tau)$ is a sample average computed on-the-fly using a single run of an integrator. Define $A^h(\tau)$ as the limit as the number of samples tends to ∞:

$$A^h(\tau) = \lim_{N\to\infty} \left(\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{P}^T_{\lfloor (\tau+t_k)/h \rfloor} \boldsymbol{P}_{\lfloor t_k/h \rfloor} \right). \tag{8}$$

The difficulty is that this limit does not generally exist if the integrator is explicit.

This paper extends a new strategy to solve the SDE (6) that combines an explicit integrator with Monte-Carlo sampling methods [8]. The resulting 'Metropolized integrator' not only preserves the equilibrium distribution of the SDE, but is often provably ergodic. This feature motivates the use of Metropolized integrators as sampling methods [2,11,38,42,45]. The idea of using the Metropolis–Hastings criterion as a numerical stabilization device seems to appear for the first time in [1]. The paper [8] proves that a Metropolized integrator approximates pathwise the SDE's solution on finite-time intervals.

A main point of this paper is that these properties ensure that a Metropolized integrator can be used to estimate dynamics along an infinitely long solution of (6). Indeed one can generate a long-time trajectory of a Metropolized integrator, and along any finite-time interval update sample averages of dynamic quantities. These sample averages converge as a consequence of ergodicity of a Metropolized integrator. The averages can also be made arbitrarily close to the true solution's average by selecting the time-step small enough. For example, a Metropolized integrator can be used to approximate to arbitrary precision the equilibrium correlation function $A(\tau)$. In fact, we show in this paper for every $T > 0$, there exists a $C(T) > 0$ such that for h sufficiently small

$$\sup_{\tau\in[0,T]} |A^h(\tau) - A(h\lfloor \tau/h \rfloor)| \leqslant C(T)h. \tag{9}$$

Since the constant $C(T)$ typically increases with the length of the time-span T, one cannot use this integrator in situations where T is very large like rare event simulation. For such problems we refer the reader to methods for molecular systems with rare events such as milestoning [51,52].

The error estimate (9) provides a theoretical order of accuracy of a Metropolized integrator. However, several questions remain:

- to what extent is the patch scalable with respect to system size?
- does the patch work for molecular systems with holonomic constraints?
- does the patch work for integators with multiple time-steps?

A main aim of this paper is to answer these questions. The paper is organized as follows:

Section 2 provides step-by-step instructions on how to patch explicit integrators based on Verlet, RATTLE, and RESPA, and some basic theory explaining why the patch works; item Section 3 conducts numerical tests on a Lennard–Jones fluid and 'dumbbell' systems;
Section 4 contains some conclusions.

2. Patch

The patch involves splicing an explicit integrator with Metropolis–Hastings steps. The 'patched integrator' possesses the following properties:

(**P1**) preservation of the SDE's equilibrium distribution; and,
(**P2**) pathwise accuracy on finite time-intervals.

In what follows we patch an explicit integrator for the SDE based on splitting (6) into Hamilton's equations:

$$\begin{cases} \frac{d\boldsymbol{Q}}{dt} = \boldsymbol{M}^{-1}\boldsymbol{P}, \\ \frac{d\boldsymbol{P}}{dt} = -\nabla U(\boldsymbol{Q}) \end{cases} \tag{10}$$

and equations describing the effect of the thermostat:

$$\begin{cases} \frac{d\boldsymbol{Q}}{dt} = 0, \\ d\boldsymbol{P} = d\boldsymbol{\eta}. \end{cases} \tag{11}$$

Define the explicit integrator as a composition of a step of Verlet for (10) and a step of an approximation to (11) (or vice versa). Replacing Verlet with an implicit method (e.g. implicit Euler) yields a scheme that satisfies property (P2). However, due to discretization error, it does not satisfy property (P1) [46–48].

Before we continue let us introduce some notation. Let $\Omega = \mathbb{T}^v \times \mathbb{R}^v$ denote the phase space of the molecular system and $\boldsymbol{Y}(t) = (\boldsymbol{Q}(t), \boldsymbol{P}(t)) \in \Omega$ denote the true solution of (6) at time $t \geqslant 0$ with initial condition $\boldsymbol{Y}(0) = \boldsymbol{x}$.

2.1. Explicit Integrator

Given a time-step h and time interval T, set the number of steps to be $N = \lfloor T/h \rfloor$ and introduce an evenly-spaced mesh in time $t_k = hk$ for all $0 \leqslant k \leqslant N$. Let $\psi_{t_{k+1},t_k} : \Omega \to \Omega$ denote an approximation to the thermostat dynamics (11). This map depends on time because of the stochastic effects in the thermostat. For example, for Langevin dynamics the thermostat force is given by,

$$d\boldsymbol{\eta} = -\gamma \boldsymbol{M}^{-1}\boldsymbol{P}\,dt + \sqrt{2\gamma\beta^{-1}}\,d\boldsymbol{W},$$

where $\boldsymbol{W}$ is a v-dimensional Wiener process and γ is a thermostat parameter [9,43]. In this case the solution to (11) is given by:

$$\psi_{t_{k+1},t_k} : (\boldsymbol{q},\boldsymbol{p}) \mapsto (\boldsymbol{q}, e^{-\gamma \boldsymbol{M}^{-1} h}\boldsymbol{p} + \boldsymbol{\eta}_{t_{k+1},t_k}), \tag{12}$$

where we have introduced the random vector:

$$\boldsymbol{\eta}_{t_{k+1},t_k} = \sqrt{2\gamma\beta^{-1}} \int_{t_k}^{t_{k+1}} e^{-\gamma \boldsymbol{M}^{-1}(t_{k+1}-s)}\,d\boldsymbol{W}(s).$$

The map ψ_{t_{k+1},t_k} satisfies property (P1). For other SDE-based thermostats an approximate map that satisfies (P1) can be similarly constructed. Notice that this map does not alter the configuration of the molecular system.

We introduce a second map $\hat{\theta}_h : \Omega \to \Omega$ which approximates (10). Since the Hamiltonian is time-independent, this map depends only on the time-step. The patch below requires that this map is symmetric and volume-preserving [20,28]. A splitting scheme for (6) is then given by:

$$\hat{\phi}_{t_{k+1},t_k} = \psi_{t_{k+1},t_k} \circ \hat{\theta}_h. \tag{13}$$

The order in (13) does not matter since the integrator's single step accuracy is $\mathcal{O}(h^2)$ either way. To improve scalability the map $\hat{\theta}_h$ consists of a composition of Verlet moves that update different parts of the molecular system as follows.

Split the molecular system into m sets of particles so that each set has v_j degrees of freedom. Given a time-step h and initial condition $\boldsymbol{x} \in \Omega$, define a single step of (13) as:

$$\hat{\boldsymbol{X}}_1 = \psi_{t_1,t_0} \circ \underbrace{\hat{\theta}_{h,m} \circ \cdots \circ \hat{\theta}_{h,1}}_{\hat{\theta}_h}(\boldsymbol{x}). \tag{14}$$

This update outputs a numerical approximation to $\boldsymbol{Y}(h)$, $\hat{\boldsymbol{X}}_1 \approx \boldsymbol{Y}(h)$. Define each map $\hat{\theta}_{h,j}$ as a Verlet update of the position and momentum of the jth set of molecules fixing all other degrees of freedom. More precisely given an input

$$\boldsymbol{x}_0 = ((\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m),(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_m)),$$

the map $\hat{\theta}_{h,j}$ outputs

$$\hat{\theta}_{h,j}(\boldsymbol{x}_0) = ((\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\hat{\boldsymbol{q}}_j,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m),(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_{j-1},\hat{\boldsymbol{p}}_j,\boldsymbol{p}_{j+1},\ldots,\boldsymbol{p}_m)).$$

Here:

$$\begin{cases} \hat{\boldsymbol{p}}_{j,\frac{1}{2}} = \boldsymbol{p}_j - \frac{h}{2}\nabla_{Q_j} U(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m), \\ \hat{\boldsymbol{q}}_j = \boldsymbol{q}_j + h\boldsymbol{M}_j^{-1}\hat{\boldsymbol{p}}_{j,\frac{1}{2}}, \\ \hat{\boldsymbol{p}}_j = \hat{\boldsymbol{p}}_{j,\frac{1}{2}} - \frac{h}{2}\nabla_{Q_j} U(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\hat{\boldsymbol{q}}_j,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m), \end{cases} \tag{15}$$

where $\boldsymbol{M}_j$ is the subset of $\boldsymbol{M}$ associated to the jth set of particles.

Remark 2.1. The splitting integrator $\hat{\theta}_h$ is a generalization of the J-splitting method introduced in [15]. Constructing this method requires writing a Hamiltonian vector field as a sum of two-dimensional Hamiltonian vector fields. A J-splitting is then given by composing the flow of each of these integrable systems. By construction, the method is energy and volume-preserving. However, it is not momentum-preserving, in general.

Remark 2.2. The splitting integrator $\hat{\theta}_h$ is a first-order composition of Verlet moves. Each of these Verlet moves is defined with respect to the Hamiltonian of the full system with some degrees of freedom held fixed. Hence, each of these moves is symmetric and volume-preserving on the same space Ω. Even though the map $\hat{\theta}_h$ is first-order accurate, its single step error in preserving energy is $O(h^3)$ because each of its Verlet updates is second-order accurate.

Remark 2.3. The splitting integrator $\hat{\theta}_h$ is only conditionally stable. The time-step restriction for the jth Verlet move is $h\sqrt{L_j} < 2$, where L_j is the Lipschitz constant of the Hamiltonian vector field obtained by freezing all degrees of freedom other than those associated to the jth set of particles. By the triangle inequality, it is clear that the maximum of these Lipschitz constants is a Lipschitz constant of the original Hamiltonian vector field. Hence, the time-step restriction for the splitting is no better than that of a traditional Verlet move.

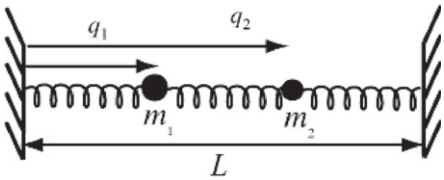
Example 2.4. Consider coupled linear springs as shown in Fig. 4(a) whose dynamics are governed by the following SDE:

$$d\boldsymbol{Z} = A\boldsymbol{Z}dt + \boldsymbol{e_4}\,dt + \sqrt{2}\boldsymbol{e_3}\,dW_1 + \sqrt{2}\boldsymbol{e_4}\,dW_2, \quad \boldsymbol{Z}(0) \in \mathbb{R}^4, \tag{16}$$

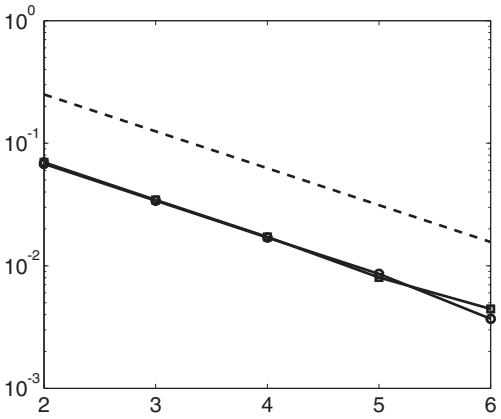
where in components $Z(t) = (q_1(t), q_2(t), p_1(t), p_2(t))$, $e_3 = (0,0,1,0)$ and $e_4 = (0,0,0,1)$; W_1 and W_2 are independent Wiener processes; and

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 1 & -1 & 0 \\ 1 & -2 & 0 & -1 \end{bmatrix}.$$

This SDE is a Langevin equation with respect to a Hamiltonian given by:



(a) Coupled Spring System



(b) Weak Accuracy

Fig. 4. SDE for coupled linear springs. (a) shows the coupled spring system, and (b) plots the weak accuracy of the approximation (14) to the SDE (16) with $\hat{\theta}_h$ respectively given by a single Verlet move for both masses (circles) and a separate Verlet move for each mass (squares). The time steps are $h = 2^{-n}$ for n as shown on the x-axis. The error is the estimate of $\mathbb{E}(q_1(1)^2 + q_2(1)^2 + p_1(1)^2 + p_2(1)^2) = 2.27661554198014615\ldots$ computed by ensemble averaging over 2×10^7 realizations. The initial condition is the springs at rest with $q_1(0) = 1/3$ and $q_2(0) = 2/3$. The dashed curve is the graph of $2^{-n}(=h)$.

$$H(q_1,q_2,p_1,p_2)=\frac{1}{2m_1}p_1^2+\frac{1}{2m_2}p_2^2+\frac{k_1}{2}q_1^2+\frac{k_2}{2}(q_2-q_1)^2+\frac{k_3}{2}(q_2-L)^2,$$

with all parameters set equal to one, for simplicity. The weak accuracy of the approximation (14) is first-order as shown in Fig. 4(b) with $\hat{\theta}_h$ respectively given by a Verlet move of both masses and a separate Verlet move for each mass.

2.2. Metropolized integrator

To derive an integrator that satisfies property (P1), we patch the explicit integrator (14) with Metropolis accept or reject steps as follows. Given a time-step h, an initial condition $\boldsymbol{x}\in\Omega$, and uniform random numbers $\zeta_j\sim U(0,1)$ for $1\leqslant j\leqslant m$, one step of a Metropolized Verlet integrator determines $\boldsymbol{X}_1\approx\boldsymbol{Y}(h)$ using:

$$\boldsymbol{X}_1=\psi_{t_1,t_0}\circ\underbrace{\theta_{h,m}\circ\cdots\circ\theta_{h,1}}_{\theta_h}(\boldsymbol{x}). \tag{17}$$

Here we have introduced the stochastic maps $\theta_{h,j}:\Omega\to\Omega$ which are Metropolized versions of the deterministic updates $\hat{\theta}_{h,j}$. Given input

$$\boldsymbol{x}_0=((\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m),(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_m)),$$

the map $\theta_{h,j}$ computes a proposed move

$$\boldsymbol{x}_1^\star=((\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m),(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_{j-1},\boldsymbol{p}_j^\star,\boldsymbol{p}_{j+1},\ldots,\boldsymbol{p}_m)).$$

using a Verlet update

$$\begin{cases}\boldsymbol{p}_{j,\frac{1}{2}}^\star=\boldsymbol{p}_j-\frac{h}{2}\nabla_{Q_j}U(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m),\\ \boldsymbol{q}_j^\star=\boldsymbol{q}_j+h\boldsymbol{M}_j^{-1}\boldsymbol{p}_{j,\frac{1}{2}}^\star,\\ \boldsymbol{p}_j^\star=\boldsymbol{p}_{j,\frac{1}{2}}^\star-\frac{h}{2}\nabla_{Q_j}U(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m)\end{cases} \tag{18}$$

and accepts this proposed move with probability

$$1\wedge\exp\left(-\beta\left[H(\boldsymbol{x}_1^\star)-H(\boldsymbol{x}_0)\right]\right). \tag{19}$$

If this proposal is rejected the momentum is reversed. To summarize

$$\theta_{h,j}:\boldsymbol{x}_0\mapsto\boldsymbol{x}_1,$$

with output $\boldsymbol{x}_1$ defined as

$$\boldsymbol{x}_1=((\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\bar{\boldsymbol{q}}_j,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m),(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_{j-1},\bar{\boldsymbol{p}}_j,\boldsymbol{p}_{j+1},\ldots,\boldsymbol{p}_m)).$$

where

$$(\bar{\boldsymbol{q}}_j,\bar{\boldsymbol{p}}_j),=\begin{cases}\left(\boldsymbol{q}_j^\star,\boldsymbol{p}_j^\star\right), & \text{if } \zeta_j<1\wedge\exp(-\beta[H(\boldsymbol{x}^\star)-H(\boldsymbol{x})]),\\ (\boldsymbol{q}_j,-\boldsymbol{p}_j), & \text{otherwise.}\end{cases} \tag{20}$$

Remark 2.5. The patch consists of accept or reject steps like (20). Like other Monte-Carlo algorithms, calculating the acceptance probability requires an energy evaluation at the current step and at the proposed move. Since the total energy is an extensive (scale-dependent) function, the acceptance probability does not scale well if the proposed move is global. This scalability problem can always be solved by using local proposal moves in (17) because the change in energy induced by a local move is an intensive quantity. However, the computation of this local energy change may be costly if the molecular interaction is long range and decays like $1/r$. This issue will be investigating in a forthcoming paper.

Remark 2.6. To satisfy detailed balance, the momentum of the jth set of particles is reversed when a proposed move is rejected. The impact of momentum reversal on the approximation to the dynamics is investigated in [1]. While these rejections ensure the integrator preserves the SDE's equilibrium distribution, they cause an $O(1)$ error in accuracy due to momentum reversals. Nevertheless (17) remains accurate as clarified in [8].

2.3. Quantitative error estimates

Consider once more the Hamiltonian of the molecular system:

$$H(\boldsymbol{q},\boldsymbol{p})=\frac{1}{2}\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}+U(\boldsymbol{q}), \tag{21}$$

where $\boldsymbol{q} \in \mathbb{T}^v$ and $\boldsymbol{p} \in \mathbb{R}^v$ represent a configuration and momentum of the molecular system, respectively. For simplicity, we assume the thermostat is given by Langevin dynamics [9,43]:

$$\begin{cases} \frac{d\boldsymbol{Q}}{dt} = \boldsymbol{M}^{-1}\boldsymbol{P}, \\ d\boldsymbol{P} = -\nabla U(\boldsymbol{Q})\,dt - \gamma \boldsymbol{M}^{-1}\boldsymbol{P}\,dt + \sqrt{2\gamma\beta^{-1}}\,d\boldsymbol{W}, \end{cases} \tag{22}$$

where γ is a thermostat parameter, β is the 'inverse temperature' entering the distribution (5), and $\boldsymbol{W}$ is a v-dimensional Wiener process. The theory below relies on the following regularity of the potential energy.

Assumption 2.7. The potential energy $U : \mathbb{T}^v \to \mathbb{R}$ is smooth.

Remark 2.8. This assumption does not permit the potential force to have singularities. For example, it holds for Morse potential interactions, but not Lennard–Jones interactions. One can relax this requirement by replacing smoothness of U by some coercivity.

Assumption 2.9. The volume-preserving map θ_h in (17) is globally momentum-invertible.

Remark 2.10. The concept of 'momentum-invertibility' is introduced in Definition 4.1 of [44], and is commonly used as an ingredient to prove ergodicity of hybrid Monte-Carlo schemes. As shown in for instance Theorem 2.2 of [7], Assumptions 2.7 and 2.9 are *sufficient* to prove that the probability transition function of the integrator is smooth and everywhere strictly positive, and hence, that the integrator satisfies a minorization condition.

Let $\mathbb{E}^\mu$ denote expectation conditional on the initial distribution being μ:

$$\mathbb{E}^\mu(g(\boldsymbol{X}_k)) = \int_\Omega \mathbb{E}^{\boldsymbol{x}}(g(\boldsymbol{X}_k))\mu(d\boldsymbol{x}), \quad \boldsymbol{X}_0 = \boldsymbol{x}.$$

The following theorem states that the Metropolized integrator satisfies (P2).

Theorem 2.11. Assume 2.7. *For every $T > 0$, there exist positive constants h_c and $C(T)$ such that,*

$$\left(\mathbb{E}^\mu\left\{|\boldsymbol{X}_{\lfloor t/h \rfloor} - \boldsymbol{Y}(h\lfloor t/h \rfloor)|^2\right\}\right)^{1/2} \leqslant C(T)h$$

for all $h < h_c$ and $t \in [0,T]$.

A proof of this result relies on single-step accuracy and bounds on moments of the Metropolized integrator (see [8]). These bounds enable the single-step error estimate to be boosted to a global error estimate. The Metropolized integrator initiated from equilibrium satisfies such bounds as a consequence of (P1).

Theorem 2.11 does not require that the Metropolized integrator be ergodic, but only that it preserves the equilibrium distribution. For this reason one can extend this result to Metropolis-adjusted discretizations of other SDE-based thermostats. Ergodicity is technically difficult to establish when the thermostat force is a nonlinear function of the state or highly degenerate (see, e.g. [27]). However, for Langevin dynamics (22) with linear friction and additive noise on all momenta, ergodicity is straightforward to establish for the Metropolized integrator.

Theorem 2.12 (*Ergodicity*). (*Assume 2.7 and 2.9*). *For all $h > 0$, observables $g : \Omega \to \mathbb{R}$, and initial conditions $x \in \Omega$,*

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T g(\boldsymbol{X}_{\lfloor t/h \rfloor})\,dt = \int_\Omega g(\boldsymbol{x})\mu(d\boldsymbol{x}),$$

where $X_0 = x$.

Proof. This proof is terse. We prove this statement for the Metropolized integrator based on a trivial partition of the molecular system. For more details please see [8] and references therein. The Metropolized integrator by construction satisfies property (P1). Moreover, its acceptance probability is strictly less than one everywhere, and the integrator without accept or reject steps admits a smooth transition density when sampled every other step by Assume 2.7 and 2.9. Please see Remark 2.10. These two observations imply that the smooth part of the two-step transition probability of the Metropolized integrator is supported everywhere, and hence, the chain is irreducible. Irreducibility and property (P1) together imply ergodicity [35]. □

Theorems 2.11 and 2.12 are sufficient to establish that the Metropolized integrator can estimate dynamics along equilibrium trajectories of (22). For example, as a corollary to the above theorems one can prove that the Metropolized integrator can be used to compute equilibrium correlation functions.

Corollary 2.13. (*Assume 2.7 and 2.9*). *For every $T > 0$ there exist positive constants h_c and $C(T)$ such that*

$$|A^h(\tau) - A(h\lfloor \tau/h \rfloor)| \leqslant C(T)h$$

for all $h < h_c$ and $\tau \in [0,T]$.

A proof of this result is provided in Appendix A.

Remark 2.14. The strong convergence estimate in Theorem 2.11 is not necessary to prove Corollary 2.13. In fact, a weak convergence estimate suffices. However, a new multilevel Monte Carlo method that substantially reduces the computational complexity of computing finite-time statistics does necessitate the strong convergence concept [19].

2.4. Case of multiple time-steps

A widely used multi-step explicit integrator for molecular systems at constant energy is the RESPA algorithm [31,49]. It is a version of Verlet that relaxes a time-step restriction imposed by rapidly changing short-range interactions. The scheme evaluates short-range forces on a smaller time-step, and hence, more frequently than long-range forces. Its overall accuracy of the algorithm is dictated by its largest time-step. For Hamiltonian ODEs, RESPA is known to exhibit resonance instabilities as described in [17]. The resonance occurs between forces evaluated at the coarse time-step and the normal modes of the molecular system excluding long-range interactions. These numerical instabilities also appear in generalizations of RESPA to Langevin SDEs [16,25,33]. In the following we show how to patch such a generalization to control numerical resonance.

Again we partition the molecular system into m sets of particles so that each set has v_i degrees of freedom and $\sum_{i=1}^m v_i = v$. We assume that the potential energy can be decomposed into fast and slow interactions:

$$U(\boldsymbol{q}) = U_{\text{slow}}(\boldsymbol{q}) + U_{\text{fast}}(\boldsymbol{q}).$$

We further assume fast and slow potential forces are respectively evaluated at h_f and h time increments.

Define the patched integrator to be (17) except that $\theta_{h,j}$ is replaced with a Metropolized version of RESPA that separately updates each part of the molecular system. Given a small time-step h_f, large time-step h, and input

$$\boldsymbol{x}_0 = ((\boldsymbol{q}_1, \ldots, \boldsymbol{q}_m), (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_m)).$$

Set $N_f = \lfloor h/h_f \rfloor$. The map $\theta_{h,j}$ computes a proposed move

$$\boldsymbol{x}_1^\star = \left(\left(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{j-1}, \boldsymbol{q}_j^\star, \boldsymbol{q}_{j+1}, \ldots, \boldsymbol{q}_m\right), \left(\boldsymbol{p}_1, \ldots, \boldsymbol{p}_{j-1}, \boldsymbol{p}_j^\star, \boldsymbol{p}_{j+1}, \ldots, \boldsymbol{p}_m\right)\right),$$

using a RESPA update

$$\begin{cases} \boldsymbol{p}_{j,\frac{1}{N_f}} = \boldsymbol{p}_j - \frac{h}{2}\nabla_{Q_j} U_{\text{slow}}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_m), \\ \boldsymbol{q}_{j,\frac{1}{N_f}} = \boldsymbol{q}_j \\ \quad \begin{cases} \boldsymbol{P}_{j,\frac{k+1}{N_f}} = \boldsymbol{p}_{j,\frac{k}{N_f}} - \frac{h_f}{2}\nabla_{Q_j} U_{\text{fast}}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{j-1}, \boldsymbol{q}_{j,\frac{k}{N_f}}, \boldsymbol{q}_{j+1}, \ldots, \boldsymbol{q}_m), \\ \boldsymbol{q}_{j,\frac{k+1}{N_f}} = \boldsymbol{q}_{j,\frac{k}{N_f}} + h_f \boldsymbol{M}_j^{-1} \boldsymbol{P}_{j,\frac{k+1}{N_f}}, \\ \boldsymbol{p}_{j,\frac{k+1}{N_f}} = \boldsymbol{P}_{j,\frac{k+1}{N_f}} - \frac{h_f}{2}\nabla_{Q_j} U_{\text{fast}}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{j-1}, \boldsymbol{q}_{j,\frac{k+1}{N_f}}, \boldsymbol{q}_{j+1}, \ldots, \boldsymbol{q}_m), \end{cases} \\ \boldsymbol{q}_j^\star = \boldsymbol{q}_{j,1} \\ \boldsymbol{p}_j^\star = \boldsymbol{p}_{j,1} - \frac{h}{2}\nabla_{Q_j} U_{\text{slow}}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{j-1}, \boldsymbol{q}_j^\star, \boldsymbol{q}_{j+1}, \ldots, \boldsymbol{q}_m), \end{cases}$$

where the inner index k runs from $1,\ldots,N_f - 1$ (where N_f is the number of steps taken at the small time-step h_f) and the outer index j runs from $1,\ldots,m$ (where m is the number of elements in the partition). This proposed move is accepted with probability

$$1 \wedge \exp\left(-\beta\left[H(\boldsymbol{x}_1^\star) - H(\boldsymbol{x}_0)\right]\right). \tag{23}$$

If this proposal is rejected the momentum is reversed as before. This generalization of RESPA achieves a speedup whenever the slow potential force is expensive to evaluate relative to the fast potential force.

2.5. Case of holonomic constraints

Here we show how to patch integrators for molecular systems with holonomic constraints. Similar issues are treated in [21,30] from the viewpoint of sampling in the presence of constraints. Again we partition the molecular system into m sets of particles so that each set has v_i degrees of freedom. The main difference to the previous cases is the possibility that the dynamics along each element of this partition is not well-defined. For instance, if the constraint couples all atoms in the molecule, then only the trivial partition will lead to well-posed dynamics. Or if the constraint couples pairs of molecules, then only partitions in terms of these pairs are permissible.

To rule out this possibility, assume that the jth set of the partition has an associated scalar constraint function $g_i : \mathbb{T}^{\nu} \to \mathbb{R}$ independent from the positions of the other sets. The case of vectorial constraints can be handled quite similarly, but for clarity we only consider scalar constraints here. The intersection of the zero level sets of these constraint functions defines the constraint manifold $\Sigma = \cap_{i=1}^{m} g_i^{-1}(0) \subset \Omega$. The velocities of the constrained atoms are tangent to this manifold. These observations motivate introducing the set of all constrained positions and momenta denoted by $T\Sigma$:

$$T\Sigma = \left\{ (\boldsymbol{q},\boldsymbol{p}) \in \Omega | g_i(\boldsymbol{q}) = 0,\ \nabla_{\mathbf{Q}_i} g_i(\boldsymbol{q})^T \boldsymbol{M}_i^{-1} \boldsymbol{p}_i = 0,\ i = 1,\dots,m \right\},$$

where $\boldsymbol{M}_i$ is the subset of the global mass matrix $\boldsymbol{M}$ associated to the ith set of particles.

For a molecular system with constraints, the probability distribution (5) is replaced by:

$$d\mu_{T\Sigma}(\boldsymbol{q},\boldsymbol{p}) \overset{\text{def}}{=} Z^{-1} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\, d\sigma_{T\Sigma}(\boldsymbol{q},\boldsymbol{p}), \quad Z = \int_{T\Sigma} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\, d\sigma_{T\Sigma}(\boldsymbol{q},\boldsymbol{p}). \tag{24}$$

Here the measure $\sigma_{T\Sigma}$ represents standard volume measure on $T\Sigma$. Introduce the Lagrange multiplier $\lambda_i(t) \in \mathbb{R}$, and assume the stochastic dynamics of the constrained molecular system is of the form:

$$\begin{cases} \frac{d\mathbf{Q}_i}{dt} = \boldsymbol{M}_i^{-1} \boldsymbol{P}_i, \\ d\boldsymbol{P}_i = -\nabla_{\mathbf{Q}_i} U(\mathbf{Q})\, dt + \nabla_{\mathbf{Q}_i} g_i(\mathbf{Q})\, d\lambda_i + d\boldsymbol{\eta}_i(\mathbf{Q},\boldsymbol{P}), \\ g_i(\mathbf{Q}) = 0, \end{cases} \tag{25}$$

where i enumerates the elements in the partition. To check that (24) is an invariant measure of (25), it suffices to show its density is a stationary solution of the corresponding Fokker–Planck equation. As before assume the solution to (25) is ergodic with respect to the probability distribution $\mu_{T\Sigma}$. To eliminate the Lagrange multiplier appearing in (25), 'differentiate' the constraint function twice along a path of $(\mathbf{Q},\boldsymbol{P})$ following either the heuristic approach in [50] or the rigorous approach described in [6,12].

Define a constrained version of (14) by splitting (25) into a constrained Hamiltonian system:

$$\begin{cases} \frac{d\mathbf{Q}_i}{dt} = \boldsymbol{M}_i^{-1} \boldsymbol{P}_i, \\ \frac{d\boldsymbol{P}_i}{dt} = -\nabla_{\mathbf{Q}_i} U(\mathbf{Q}) + \nabla_{\mathbf{Q}_i} g_i(\mathbf{Q}) \lambda_{i,1}, \\ g_i(\mathbf{Q}) = 0 \end{cases} \tag{26}$$

and constrained thermostat dynamics:

$$\begin{cases} \frac{d\mathbf{Q}_i}{dt} = 0, \\ d\boldsymbol{P}_i = d\boldsymbol{\eta}_i + \nabla_{\mathbf{Q}_i} g_i(\mathbf{Q})\, d\lambda_{i,2}, \\ 0 = \nabla_{\mathbf{Q}_i} g_i(\mathbf{Q})^T \boldsymbol{M}_i^{-1} \boldsymbol{P}_i. \end{cases} \tag{27}$$

Approximate the solution to (26) by using a constrained version of Verlet known as RATTLE [40]. As before to improve scalability separately propagate each set of the partition using RATTLE. Since RATTLE moves are time-reversible and volume-preserving [29], a Metropolis method based on a RATTLE proposed move and the probability distribution (24) yields an acceptance probability that is a function of the change in energy induced by the separate RATTLE moves.

As before compose this map with an approximation to (27) which we denote by $\psi_{t_{k+1},t_k} : T\Sigma \to T\Sigma$. If we assume that the thermostat in (27) is given by Langevin dynamics, then its solution is given by:

$$\psi_{t_{k+1},t_k} : \{(\boldsymbol{q}_i,\boldsymbol{p}_i)\}_{i=1}^{m} \mapsto \left\{ \left(\boldsymbol{q}_i, e^{-\gamma \mathbb{P}_i(\boldsymbol{q}) \boldsymbol{M}_i^{-1} h} \boldsymbol{p}_i + \boldsymbol{\eta}^i_{t_{k+1},t_k} \right) \right\}_{i=1}^{m}, \tag{28}$$

where we have introduced the random vector:

$$\boldsymbol{\eta}^i_{t_{k+1},t_k} = \sqrt{2\gamma\beta^{-1}} \int_{t_k}^{t_{k+1}} e^{-\gamma \mathbb{P}_i(\boldsymbol{q}) \boldsymbol{M}_i^{-1} (t_{k+1}-s)} \mathbb{P}_i(\boldsymbol{q}) d\boldsymbol{W}_i(s)$$

and the projection matrix:

$$\mathbb{P}_i(\boldsymbol{q}) = \boldsymbol{I} - (\nabla_{\mathbf{Q}_i} g_i)[(\nabla_{\mathbf{Q}_i} g_i)^T \boldsymbol{M}_i^{-1} (\nabla_{\mathbf{Q}_i} g_i)]^{-1} (\nabla_{\mathbf{Q}_i} g_i)^T \boldsymbol{M}_i^{-1}.$$

Here $\boldsymbol{I}$ is the $v_i \times v_i$ identity matrix. To derive (28) eliminate the Lagrange multiplier in (27) by differentiating the momentum constraint and using as an integrating factor the matrix $\exp\left(\gamma \mathbb{P}_i(\boldsymbol{q}) \boldsymbol{M}_i^{-1} t\right)$.

Remark 2.15. When the mass matrix is not the identity, (28) requires computing the exponential of a position dependent matrix. This calculation is non-trivial, and can be avoided by using SHAKE in place of RATTLE and an unconstrained velocity update in place of (28). The resulting Metropolized integrator will satisfy property (P1) with respect to an equilibrium distribution with unconstrained velocities. It will also satisfy property (P2).

Given a time-step h, initial condition $\boldsymbol{x} \in \Omega$, and uniform random numbers $\zeta_j \sim U(0,1)$ for $j = 1,\ldots,m$, one step of the Metropolized integrator determines $\boldsymbol{X}_1$ using (17), but with the proposed moves in $\theta_{h,j}$ obtained by RATTLE as follows. Introduce the discrete Lagrange multipliers $\lambda_{j,1}, \lambda_{j,2} \in \mathbb{R}$. The integrator inputs a point in phase space on the constraint manifold

$$\boldsymbol{x}_0 = ((\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m),(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_m))$$

and outputs a proposed move on the constraint manifold

$$\boldsymbol{x}_1^\star = \Big(\Big(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m\Big),\Big(\boldsymbol{p}_1,\ldots,\boldsymbol{p}_{j-1},\boldsymbol{p}_j^\star,\boldsymbol{p}_{j+1},\ldots,\boldsymbol{p}_m\Big)\Big),$$

where $\left(\boldsymbol{q}_j^\star,\boldsymbol{p}_j^\star\right)$ are determined by

$$\begin{cases} \boldsymbol{p}_{j,\frac{1}{2}}^\star = \boldsymbol{p}_j - \frac{h}{2}\nabla_{Q_j}U(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m) - \frac{h}{2}\lambda_{j,1}\nabla_{Q_j}g_j(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m), \\ \boldsymbol{q}_j^\star = \boldsymbol{q}_j + h\boldsymbol{M}_j^{-1}\boldsymbol{p}_{j,\frac{1}{2}}^\star, \\ \boldsymbol{p}_j^\star = \boldsymbol{p}_{j,\frac{1}{2}}^\star - \frac{h}{2}\nabla_{Q_j}U(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m) \\ \qquad\quad - \frac{h}{2}\lambda_{j,2}\nabla_{Q_j}g_j(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m), \\ 0 = g_j\Big(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m\Big), \\ 0 = \nabla_{\boldsymbol{Q}_j}g_j\Big(\boldsymbol{q}_1,\ldots,\boldsymbol{q}_{j-1},\boldsymbol{q}_j^\star,\boldsymbol{q}_{j+1},\ldots,\boldsymbol{q}_m\Big)^T\boldsymbol{M}_j^{-1}\boldsymbol{p}_j^\star. \end{cases} \tag{29}$$

The Lagrange multiplier $\lambda_{j,1}$ enforces that the jth proposed positions satisfy the position constraint, and $\lambda_{j,2}$ enforces that the jth proposed velocities are tangent to the constraint manifold. This proposed move is accepted with probability

$$1 \wedge \exp\left(-\beta\left[H(\boldsymbol{x}_1^\star) - H(\boldsymbol{x}_0)\right]\right). \tag{30}$$

This acceptance probability is a function of the change in energy induced by the RATTLE proposed move. If this proposal is rejected the velocity is reversed, but remains tangent to the constraint manifold.

3. Applications

This section applies the Metropolized integrators introduced in Section 2.

3.1. Lennard–Jones fluid

A Lennard–Jones fluid consists of n identical particles with pairwise interactions given by a Lennard–Jones potential energy. In what follows we use dimensionless units to describe this system. In these units mass is rescaled by the mass of an individual particle (so that the particles have unit mass), energy by the depth of the Lennard–Jones potential energy, and length by the point where the potential energy is zero. We follow the notation and setup provided in Part I of [18].

We simulate the Lennard–Jones fluid in a fixed periodic box which we call the simulation box. We used a truncated version of the Lennard–Jones potential in which the energy between two particles a distance r apart is kept constant after a certain cutoff distance and is given by:

$$U_{LJ}(r) = \begin{cases} f(r) - f(r_c), & r < r_c, \\ 0, & \text{otherwise}, \end{cases} \tag{31}$$

where we have introduced $f(r) = 4(1/r^{12} - 1/r^6)$ and r_c is bounded above by the size of the simulation box. Other shifts can be used to make the higher derivatives of U_{LJ} continuous. The error introduced by the truncation in (31) is proportional to the density of the molecular system and can be made arbitrarily small by selecting the cutoff distance sufficiently large.

The pair potentials are a function of the distance between the ith and jth particle. If the position of the ith and jth particles are $\boldsymbol{q}_i$ and $\boldsymbol{q}_j$, and the length of the simulation box ℓ, then this distance is given by:

$$r_{i,j}(\boldsymbol{q}) = |(\boldsymbol{q}_i - \boldsymbol{q}_j) \bmod \ell|.$$

In terms of this pairwise distance, the potential energy of a Lennard–Jones fluid is a sum of interactions between all pairs of particles:

$$U(\boldsymbol{q}) = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} U_{LJ}(r_{i,j}(\boldsymbol{q})). \tag{32}$$

Evaluating the potential force requires $O(n^2)$ operations (where v is the dimension of configuration space), and typically dominates total computation cost.

3.2. Autocorrelation of Lennard–Jones fluid

Here we test the accuracy of the Metropolized integrator (17) based on trivial and per particle partitions of the molecular system. In the former proposed moves in the Metropolis steps are obtained by per particle Verlet updates, and in the latter by global Verlet updates. The numerics indicate that the order of accuracy of both methods is roughly $O(h^2)$ and that the error constant of the per particle partition is approximately an order of magnitude smaller.

To test accuracy we use the Metropolized integrator to estimate the equilibrium momentum autocorrelation of the Lennard–Jones fluid. For the numerical experiment, we set the fluid's density $\rho = 0.8442$ and temperature $\mathcal{T} = 0.728$ (units are dimensionless as described earlier). For these values the phase of the Lennard–Jones fluid is liquid, and close to the triple (gas–liquid–solid) point. We also fix the number of particles to be $n = 25$ and the degrees of freedom to be $d = 2$. The size of the simulation box is $\ell = (n/\rho)^{1/2} \approx 5.04$. A reasonable cutoff distance in (31) at the selected density is $r_c = 2.5$. The initial positions of the particles are chosen to be the vertices of a square lattice that fills the simulation box. For instance, the length of each square in the lattice can be chosen to be $\ell/\lceil n^{1/2} \rceil$. The initial velocities are sampled from the Maxwell distribution.

The effect of the thermostat parameter γ on the velocity autocorrelation function is shown in Fig. 5. In the subsequent experiments, the thermostat parameter is set at: $\gamma = 1.0$. At this value the effect of the thermostat on the autocorrelation function is mild. The thermostat parameter is set equal to $\gamma = 1.0$.

In the simulations we estimate the true velocity autocorrelation over a time-interval $[0,1]$. Set $N = \lfloor T/h \rfloor$ and introduce an evenly-spaced mesh in computational time $t_k = h\,k$ for all $0 \leqslant k \leqslant N$. Let $A^h : [0,1] \to \mathbb{R}$ denote the velocity correlation function obtained by the Metropolized integrator (17):

$$A^h(\tau) = \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} \boldsymbol{P}^T_{\lfloor(\tau+t_k)/h\rfloor} \boldsymbol{P}_{\lfloor(t_k)/h\rfloor} \quad (\tau \geqslant 0).$$

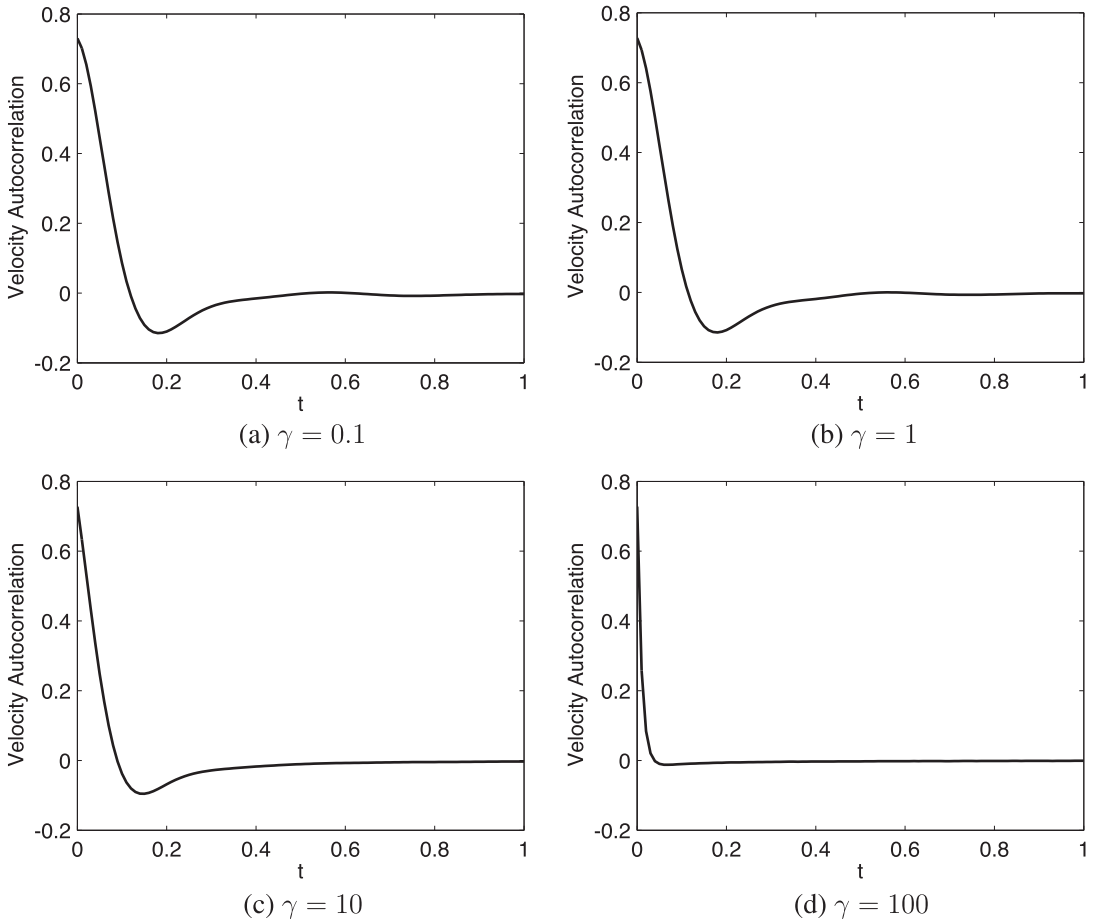


Fig. 5. Choice of thermostat parameter. This figure illustrates the effect of the damping parameter on a well-resolved approximation to the velocity autocorrelation function over the time interval $[0,1]$. The molecular system consists of 25 particles in a square box with periodic boundary conditions at a temperature of 0.728 and density 0.8442. In (c) and (d), observe that the autocorrelation function is dominated by dissipation. Whereas in (a) and (b) when γ is small, the autocorrelation function does not appear to be affected by the value of the thermostat parameter.

Define the relative Richardson error as

$$\epsilon_h \stackrel{\text{def}}{=} \frac{\sup_{\tau\in[0,1]}|A^h(\tau)-A^{2h}(\tau)|}{\sup_{\tau\in[0,1]}|A(\tau)|}. \tag{33}$$

An empirical estimate for ϵ_h is plotted for time-steps

$$h = \{0.005, 0.0025, 0.00125, 0.000625\},$$

in Fig. 6 with $N = 10^8$. The denominator in ϵ_h is calculated by using the approximation of A at $h = 0.000625$. The figure shows that the error of the Metropolized integrator is approximately $O(h^2)$. Moreover it shows that the Metropolized integrator based on a per particle partition is nearly an order of magnitude more accurate than the algorithm based on a trivial partition.

Remark 3.1. We stress that the $O(h^2)$ rate of convergence observed in Fig. 6 does not imply that the accuracy of the integrators is second-order. Indeed, the approximation of the equilibrium correlation of, for instance, the potential energy is first-order. Therefore, the estimate in Corollary 2.13 is sharp. We also stress that while large step-sizes are used in the impact oscillator example in the introduction, much smaller step-sizes are used to validate the quantitative error estimates provided in the paper.

3.3. Autocorrelation of Lennard–Jones dumbbells

A rigid dumbbell is a type of molecule that involves a holonomic constraint. It consists of a pair of particles constrained to a fixed distance from one another. The constraint arises when the spring joining a flexible dumbbell infinitely stiffens. Consider n identical dumbbells where particles in separate dumbbells interact via a Lennard–Jones potential energy (32).

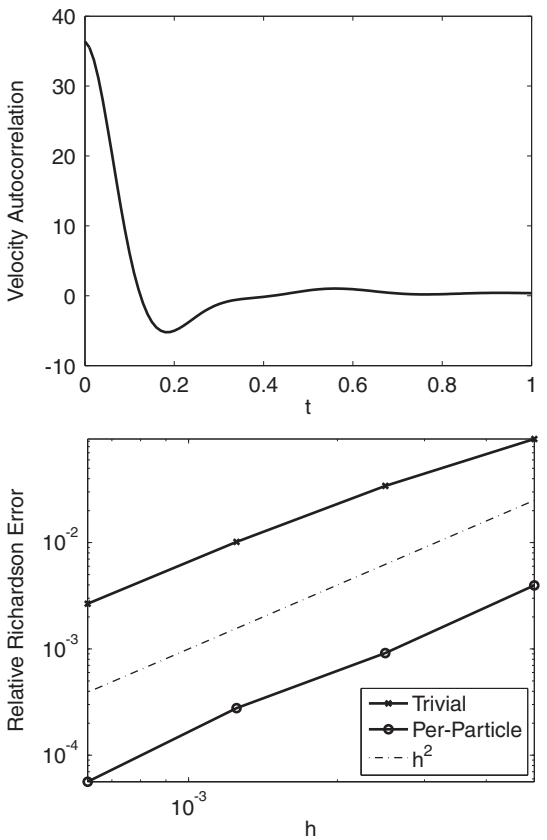


Fig. 6. Autocorrelation of Lennard–Jones fluid. The panel on the top shows the velocity autocorrelation function over the time interval [0,1]. The panel on the bottom shows a loglog plot of the Richardson error (33) of the Metropolized integrator based on trivial and per-particle partitions. The molecular system integrated is the Lennard–Jones fluid with 25 particles in a square box. The dashed line represents a reference slope of h^2. A total of $N = 10^8$ samples were used to obtain these empirical estimates. Notice that the Metropolized integrator based on per-particle Verlet moves is about an order of magnitude more accurate than the Metropolized integrator based on global Verlet moves.

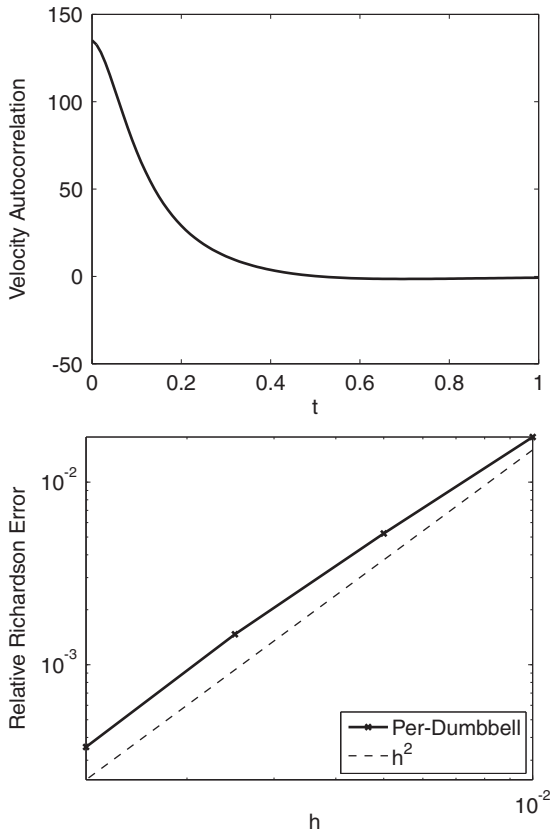


Fig. 7. Autocorrelation of Lennard–Jones dumbbells. The panel on the top shows the velocity autocorrelation function. The panel on the bottom shows a loglog plot of an empirical estimate of the Richardson error (33) of the Metropolized integrator based on per-dumbbell RATTLE proposed moves. The molecular system integrated is the Lennard–Jones system of 15 dumbbells in a square box. The dashed line represents a reference slope of h^2. A total of $N = 10^8$ samples were used to obtain this estimate.

For the numerical experiment, we simulate the dumbbells in a simulation box of length ℓ and $d = 2$ dimensions as before. We set the system's density $\rho = 0.998$ and temperature $\mathcal{T} = 3.0$ (units are dimensionless as described earlier). We also fix the number of dumbbells to $n = 30$ and the length of each dumbbell to $\ell_0 = 1$. If the positions of the ith pair of particles describing the ith dumbbell are $\boldsymbol{q}_{i,1}$ and $\boldsymbol{q}_{i,2}$, then the constraint function associated to the ith dumbbell is given by:

$$g_i(\boldsymbol{q}) = |(\boldsymbol{q}_{i,1} - \boldsymbol{q}_{i,2}) \bmod \ell|^2 - \ell_0^2, \quad i = 1, \ldots, n.$$

The size of the simulation box is $\ell = (2n/\rho)^{1/2} \approx 7$. The cutoff distance in (31) is set at $r_c = 3.0$. The initial positions of the dumbbells are chosen randomly in the simulation box, but with no overlap. The initial velocities are sampled from the Maxwell distribution constrained to be tangent to the constraint manifold at the initial positions and temperature.

To quantify the accuracy of the numerical method, we use the Metropolized integrator based on a per dumbbell partition. We again use the relative Richardson error defined in (33) to estimate the rate of convergence of the method. Fig. 7 shows the velocity autocorrelation that we wish to approximate over the time-interval [0,1]. It is noticeably different from the velocity autocorrelation in the case of the Lennard–Jones cluster without constraints. The figure also shows a plot of the Richardson error as a function of time-step. The rate of convergence appears to be approximately $O(h^2)$.

4. Conclusions

This paper extended ideas in [8] to SDEs arising in molecular dynamics. Monte Carlo methods (including Metropolis–Hastings, smart, hybrid and generalized hybrid Monte-Carlo algorithms [2,13,22,24,26,36,39]) were applied to stably and explicitly solve for the long-time dynamics of molecular systems at constant temperature governed by SDEs. The strengths and weaknesses of this approach to solving SDEs were addressed. While the paper focused on Langevin dynamics, the ideas can be extended to other SDE-based thermostats.

An open question about the Metropolized integrator is its convergence rate to equilibrium. In general, a discretization of an SDE (even a Metropolized one) does not inherit all of the mixing properties of the true solution [34]. This issue has been

studied in the overdamped limit of Langevin dynamics for the so-called Metropolis adjusted Langevin algorithm (MALA) [5]. It turns out that MALA converges to equilibrium at an exponential rate up to terms exponentially small in time-step. However, in the overdamped limit positions are no longer differentiable, and MALA does not involve momentum flips. One expects momentum flips to reduce stagnation at high energy values, and hence, enhance mixing.

Acknowledgements

We wish to thank Ben Leimkuhler for pointing us to Ref. [15], and Andrew Stuart for helpful comments.

Appendix A

Proof of Corollary 2.13. We prove this estimate for the equilibrium correlation of any Lipschitz function $g \in C^{0,1}(\Omega, \mathbb{R})$. This assume includes the special case of a scalar velocity autocorrelation function. Ergodicity of the Metropolized integrator (see Theorem 2.12) implies that

$$\lim_{N\to\infty} \frac{1}{N}\sum_{k=1}^{N} g(\boldsymbol{X}_{\lfloor t_k/h\rfloor}) g(\boldsymbol{X}_{\lfloor (t_k+\tau)/h\rfloor}) = \langle g(\boldsymbol{X}_{\lfloor t/h\rfloor}) g(\boldsymbol{X}_{\lfloor (t+\tau)/h\rfloor})\rangle$$

for any $t \geqslant 0$. (Recall, that the angle brackets denote a double average with respect to initial conditions distributed according to the equlibrium distribution of (22) and realizations of the Wiener process.) Thus, we wish to estimate:

$$\epsilon_h \stackrel{\text{def}}{=} |\langle g(\boldsymbol{X}_{\lfloor t/h\rfloor}) g(\boldsymbol{X}_{\lfloor (t+\tau)/h\rfloor})\rangle - \langle g(\boldsymbol{Y}(t)) g(\boldsymbol{Y}(t+h\lfloor \tau/h\rfloor))\rangle| = |\langle g(\boldsymbol{x})(g(\boldsymbol{X}_{\lfloor \tau/h\rfloor}) - g(\boldsymbol{Y}(h\lfloor \tau/h\rfloor)))\rangle|.$$

By the Cauchy–Schwarz inequality,

$$\epsilon_h \leqslant C_g \left(\int_\Omega |\mathbb{E}^{\boldsymbol{x}}\{g(\boldsymbol{X}_{\lfloor \tau/h\rfloor}) - g(\boldsymbol{Y}(h\lfloor \tau/h\rfloor))\}|^2 \, d\mu \right)^{1/2},$$

where $C_g = \left(\int_\Omega |g(\boldsymbol{x})|^2 d\mu \right)^{1/2}$. By Jensen's inequality,

$$\epsilon_h \leqslant C_g (\mathbb{E}^{\mu}\{|g(\boldsymbol{X}_{\lfloor \tau/h\rfloor}) - g(\boldsymbol{Y}(h\lfloor \tau/h\rfloor))|^2\})^{1/2}.$$

The Lipschitz assume on g implies,

$$|g(\boldsymbol{x}) - g(\boldsymbol{y})| \leqslant L_g |\boldsymbol{x} - \boldsymbol{y}|$$

for some constant $L_g > 0$ and for every $x, y \in \Omega$. Hence,

$$\epsilon_h \leqslant L_g C_g (\mathbb{E}^{\mu}\{|\boldsymbol{X}_{\lfloor \tau/h\rfloor} - \boldsymbol{Y}(h\lfloor \tau/h\rfloor)|^2\})^{1/2}.$$

Pathwise convergence of the Metropolized integrator (see Theorem 2.11) implies

$$(\mathbb{E}^{\mu}\{|\boldsymbol{X}_{\lfloor \tau/h\rfloor} - \boldsymbol{Y}(h\lfloor \tau/h\rfloor)|^2\})^{1/2} \leqslant C(T)h.$$

From which it follows, that the accuracy of the Metropolized integrator in computing the equilibrium correlation in g is $O(h)$ with a prefactor that depends on the function g. □

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